Tutorial — Poora Project Zero Se

Ye document maan ke chalta hai ki tumhe is project ke baare mein **kuch nahi pata**. Har term explain kiya gaya hai, har command ka asli output diya gaya hai, aur har concept ke saath ye bhi likha hai ki wo **kis file mein** hai.

`README.md` judges ke liye technical reference hai. **Ye document tumhare liye hai** — samajhne ke liye.

Contents

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1. Network traffic hoti kya hai

Jab bhi tumhara computer kisi doosre computer se baat karta hai — YouTube khola, WhatsApp bheja, app ne update check kiya — ek **connection** banta hai.

Har connection ka record bacha rehta hai, jise **flow** kehte hain. Socho phone call record:

Phone call record	Network flow
Kisne call kiya	Source IP — `147.32.84.165`
Kisko kiya	Destination IP — `74.125.232.195`
Kitni der	Duration — `2.3 seconds`
Kab	Timestamp
—	Kitna data gaya (bytes, packets)
—	Kaunse **port** pe (80 = website, 53 = DNS, 25 = email)

Port kya hai? Ek computer pe kai services chal sakti hain. Port number batata hai ki baat kis service se ho rahi hai — jaise ek building mein flat number.

Ek normal office network mein **lakhon flows per hour** bante hain. Humare dataset mein ek capture = **28 lakh flows**, 6 ghante ka.

2. Problem kya hai

Ek company ke network mein 200 computers hain. Unme se ek **hack ho chuka hai**. Security team ko pata karna hai: **kaunsa?**

Purana tareeka: ek tool har flow dekhta hai aur bolta hai "safe" ya "khatarnak".

Iski problem: attack ek flow nahi hota. Attack ek **process** hota hai jo waqt ke saath unfold hota hai:

```
Step 1: Scan          "kaunse ports khule hain?"
Step 2: Andar ghusna   weak point exploit karo
Step 3: Phone home     "main andar hu, order do"
Step 4: Failna         doosre computers pe
Step 5: Data churana   bahar bhejna
```

Ek-ek flow dekhna aisa hai jaise **poori film ko ek frame se judge karna**. Har frame akela normal lag sakta hai; sequence batati hai ki kahani kya hai.

Aur asli dikkat: purana tool tab batata hai jab **nuksaan ho chuka** hota hai.

> NTRO ka problem statement yahi maangta hai — "*predict the likelihood and progression of malicious activity before compromise is completed*"

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3. Data kahan se aaya

CTU-13 — Czech Technical University ne 2011 mein banaya. Unhone:

1. Asli computers liye
2. Unme **asli malware** daala — Neris, Rbot, Virut (ye asli botnets hain)
3. 6 se 66 ghante tak **saara network traffic record** kiya
4. Har flow pe label lagaya — normal hai ya malware ka

Botnet kya hai? Malware jo tumhare PC ko *remote-controlled zombie* bana deta hai. Wo attacker ke server ko call karta hai ("order do"), phir kaam karta hai — spam bhejta hai, fake ad clicks karta hai.

Ye data free aur legal hai (CC BY licence), aur PS mein naam se mention hai.

Humne **7 captures** use kiye → **95,518 network states** bane.

Download karne ka command section 10 mein hai.

4. Humne kya banaya — ek line mein

> **Network ka chhota simulator, jo seekhta hai ki traffic kaise badalta hai, aur usse aage chala ke batata hai ki aage kya hone wala hai.**

Weather wali misaal se samjho:

Thermometer	Weather forecast model
"abhi 35°C hai"	"3 ghante mein baarish hogi"
Sirf **abhi** batata hai	Seekhta hai ki **mausam kaise badalta hai**, phir aage chalata hai
= purana IDS	= **humara world model**

"World model" ka matlab yahi hai — AI jo duniya ka *internal simulation* banata hai. PS mein exactly yahi maanga gaya hai.

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5. Andar kya hota hai — 4 steps

Step 1 — Traffic ko chunks mein todo

Lakhon flows seedha model mein nahi daal sakte. Isliye **1-minute ke tukde** karte hain.

Har minute mein, har computer ka ek **summary** banate hain — **39 numbers**:

Number	Kya batata hai
`n_flows`	Kitne connections banaye
`n_unique_dst`	Kitne alag computers se baat ki
`n_unique_dport`	Kitne alag ports try kiye
`frac_syn_only`	Kitne connections ka **jawab hi nahi aaya** ← scanning ki nishani
`egress_ratio`	Data zyada bahar gaya ya andar aaya
`beacon_regularity`	Connections regular interval pe the ya random ← robot vs insaan
...aur 33 aur	

Ye 39 numbers ka set = **ek network state**. Matlab: "*us ek minute mein wo computer kya kar raha tha.*"

> **Asli finding:** benign computers pe `frac_syn_only` ka average **0.04** tha, infected pe **0.57**. **13 guna farak.** Malware chupke se scan kar raha tha.

📁 Code: `src/features/flow.py` (flows padhta hai), `src/features/windows.py` (states banata hai)

Step 2 — Model transitions seekhta hai

Ab model ko states ki **sequence** dikhate hain:

```
minute 1 → minute 2 → minute 3 → minute 4 → ...
```

Model sirf **ek cheez** seekhta hai:

> "Aise state ke baad, aam taur pe aisa state aata hai."

Bas. Yehi core hai — bilkul jaise weather model seekhta hai "aisa pressure pattern ho to kal baarish".

📁 Code: `src/model/world_model.py`

Step 3 — Ab traffic dikhana band kar do ← **yahi asli trick hai**

Kyunki model ne transition seekh liya hai, wo **khud se agla state bana sakta hai**. Phir usse agla. Phir usse agla. **10 baar.**

Ye 10 states **kabhi hue hi nahi** — model ne inhe *imagine* kiya hai.

Yahi forecast hai. Aur yahi world model ko normal classifier se alag karta hai.

📁 Code: `WorldModel.imagine()` — `src/model/world_model.py`

Step 4 — Har imagined state pe sawal poochho

Model ke do "heads" (chhote decision-makers) hain. Har imagined state pe:

1. **"Khatra hai?"** → 0 se 1 ke beech probability
2. **"Kill chain ke kis stage pe hai?"** → 5 stages mein se ek

Kyunki 10 baar imagine karte hain, hume **agle 10 minute ka forecast** milta hai.

Aur ye poora kaam **32 baar** karte hain (har baar thoda alag), taaki pata chale model kitna sure hai. Isse dashboard pe wo **shaded band** banta hai — band chauda = model confident nahi.

📁 Code: `src/inference.py`

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6. MITRE ATT&CK stages

MITRE ek American non-profit hai jisne attackers ke behaviour ki **standard dictionary** banayi. Duniya bhar ki security teams yahi vocabulary use karti hain.

PS ne 5 stages maange:

Stage	Matlab	Rozmarra ki misaal
Reconnaissance	Dekh raha hai kya khula hai	Chor ghar ke bahar khidkiyan check kar raha
Initial Access	Andar ghus gaya	Khidki tod ke andar
Lateral Movement	Network mein fail raha hai	Ek kamre se doosre kamre mein
Command & Control	Boss server se orders le raha	Chor phone pe gang leader se baat kar raha
Exfiltration	Data bahar bhej raha hai	Saaman bag mein bhar ke nikaalna

Stages nikaale kaise?

Ye achha wala part hai. CTU-13 ke labels sirf "botnet/normal" nahi hain — usme **detail** hai:

```
flow=From-Botnet-V42-TCP-CC16-HTTP          ← "CC" = Command & Control
flow=From-Botnet-V42-TCP-Attempt-SPAM       ← spam bhejne ki koshish
flow=From-Botnet-V49-...-Binary-Download    ← payload download
flow=From-Botnet-V42-UDP-DNS                ← DNS lookup
```

To humne stages **apne guess se nahi**, **dataset ke apne annotations se** nikaale. Judges ke saamne ye defend karna bahut aasan hai.

Asli result — infected computer ka timeline:

```
1 1 1 1 1 1 1 1 1 5 5 5 5 5 5 5 5 5 5 5 5 5 ...
└── Reconnaissance ──┘└──── Exfiltration ───┘
```

Pehle 9 minute scanning, phir spam bhejna shuru. **Ye asli transition hai** — aur model ko yahi pehle se predict karna hai.

📁 Code: `src/mitre.py` — mapping table `_LABEL_STAGE_RULES` mein hai

7. Explainability — "kyun bola?"

PS clearly kehta hai: "*Black-box outputs without interpretability are not acceptable.*"

Matlab sirf "khatra 81%" bolna kaafi nahi. **Kyun** batana padega. Humne 3 tareeke lagaye:

1. Kaunse numbers ne score badhaya

Model ko thoda-thoda karke input dete hain aur dekhte hain score kaise badalta hai. Isse har feature ka **hissa** nikalta hai.

2. Kaunse purane minute matter kiye

Model ke andar **attention** hai — wo dekhta hai ki decision lete waqt kis purane minute pe dhyan diya. Wo weights hum dikhate hain.

3. Traffic mein kya badlega ← sabse kaam ka

Model sirf risk score nahi, **agla traffic bhi predict** karta hai. To hum bol sakte hain:

> "Distinct destination ports 12 se 240 ho jaayenge"

Ye ek analyst ke liye "risk 0.81" se hazaar guna zyada useful hai — kyunki uske upar **action** liya ja sakta hai.

📁 Code: `src/explain.py`

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8. Kaise pata ki kaam karta hai

Cheating kaise roki

Agar model ko wahi data test mein do jo training mein diya tha, wo **ratt ke** 100% score le lega. Isliye:

Time se split kiya. Har capture ka:

```
|<--------- pehla 70% --------->|<- 15% ->|<- aakhri 15% ->|
         TRAINING                   VAL          TEST
```

Model ne test wala time **kabhi dekha hi nahi**. Ye asli deployment jaisa hai — aaj ka data seekho, kal ka predict karo.

Beech mein ek **guard band** bhi rakha hai taaki koi sequence boundary cross na kare.

Comparison — baselines

Sirf "hamara model achha hai" bolna bekaar hai. **Kiske comparison mein?**

Humne ek purana simple tareeka (**logistic regression**) banaya aur usse **bilkul wahi data, wahi features** diye. Fair fight.

Model	F1	Galat alarm rate (FPR)
Humara world model	**0.984**	**0.000**
Purana tareeka (8 min history)	0.744	0.004
Purana tareeka (1 min)	0.550	0.009

F1 kya hai? 0 se 1 ke beech score, jo do cheezein milata hai:

- **Precision** — jab alarm bajaya, kitni baar sach mein attack tha
- **Recall** — jitne asli attacks the, unme se kitne pakde

0.984 matlab dono lagbhag perfect. Aur **FPR 0.000** = ek bhi galat alarm nahi. SOC team ke liye ye critical hai — jhoothe alarms se log tool hi band kar dete hain.

📁 Code: `src/evaluate.py` · Result: `artifacts/reports/benchmark.md`

9. Teen honest findings

Finding 1 — Humne apne hi code mein cheating pakdi

Jo PCAP file humne download ki, usme **sirf infected computer ka traffic** tha. Model ke liye shortcut ban gaya:

> "Agar packet data available hai → ye infected hai"

Model ne behaviour seekhna chhod ke **ye shortcut ratt liya**. Proof:

	Training ke windows	Baad wale windows
Model ka score	**0.9995**	**0.003**

Cliff bilkul us line pe tha jahan training khatam hoti hai — matlab kuch seekha hi nahi tha.

Shortcut hataya → **F1 0.23 se 0.98**. Leak model ko *behtar* nahi, **kharab** kar raha tha.

> Presentation slide 4 pe ye sabse upar hai. **Apne hi kaam mein bug pakadna kisi bhi metric se strong signal hai** — zyadatar teams ye kar hi nahi paatin.

Finding 2 — Dataset mein "attack se pehle" ka data hai hi nahi

Humne check kiya: **856 malicious windows, aur infected computers ka ek bhi clean window nahi.**

Kyun? Researchers ne malware chala ke recording start ki. To "pehle normal tha, phir infected hua" wala transition **capture hi nahi hua**.

Isliye humne **kill-chain progression** ko main task banaya — usme 138 asli transitions hain.

Finding 3 — Ek signal ulta nikla

Humne socha tha: jo traffic model ko **anokha** lage, wo attack hoga. Ulta nikla:

	Average "surprise"
Normal traffic	0.49
Malware traffic	**0.35**

Malware ka traffic normal se **kam** anokha hai. Logic: **bot machine hai** — regular interval pe kaam karta hai. Insaan random hote hain.

Signal useful hai, bas **ulta padhna** padta hai. Dashboard pe wahi likha hai.

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10. Khud chala ke dekho

Setup (ek baar)

```
cd "C:\Users\Admin\Documents\SIH 2026"
python -m venv .venv
.venv\Scripts\python.exe -m pip install -r requirements.txt
```

Step A — Data download

```
curl -L -o data/raw/CTU-13-Dataset.tar.bz2 ^
  https://mcfp.felk.cvut.cz/publicDatasets/CTU-13-Dataset/CTU-13-Dataset.tar.bz2
tar -xjf data/raw/CTU-13-Dataset.tar.bz2 -C data/raw
```

~1.9 GB. Ho chuka hai to skip karo.

Step B — Traffic ko states mein badlo

```
.venv\Scripts\python.exe -m src.prepare_data
```

Asli output:

```
reading capture20110810.binetflow
capture20110810.binetflow: 2824636 flows, 1.5% botnet, span 367.2 min
state cells: 21866 over 275 hosts, 1.30% malicious
stages: 284/284 malicious cells from dataset labels, 0 from behaviour
forecast targets: 1.29% of cells precede an infiltration within 5 windows
scenario 1 done in 75.6s
```

Padho kaise: 28 lakh flows → 21,866 states, 275 computers, 367 minute ka data. Saare 284 malicious states ke stages **dataset ke labels se** aaye (humare guess se nahi).

Step C — Model train karo

```
.venv\Scripts\python.exe -m src.train
```

Asli output:

```
temporal split: train 61522 / val 8876 / test 14776 cells
sequences: 12955 of length 16 over 571 hosts, 2.36% positive steps
world model: 138,398 parameters
epoch  1/40 (169s) train loss 10.2502 | val AP 0.3373 stage-F1 0.3520
epoch 12/40 (14s)  train loss  2.9982 | val AP 1.0000 stage-F1 0.5469
  -> new best, saved world_model.pt
done. best epoch 12
```

Padho kaise: loss neeche jaana chahiye (10.2 → 2.9 ✓), val score upar (0.33 → 1.00 ✓). Epoch 12 sabse achha tha, wahi save hua.

Step D — Benchmark chalao

```
.venv\Scripts\python.exe -m src.evaluate
```

Result yahan aata hai: `artifacts/reports/benchmark.md`

Step E — Tests

```
.venv\Scripts\python.exe -m tests.smoke_model
.venv\Scripts\python.exe -m tests.test_packet
```

Asli output:

```
parameters: 141,365
observe: shapes OK
imagine: shapes OK
imagine: no encoder dependency OK     ← ye sabse important line hai
attention: causal mask OK
all smoke checks passed
```

> **"no encoder dependency OK"** ka matlab: test ne verify kiya ki jab model imagine karta hai, tab wo **asli traffic ko chhoo bhi nahi raha**. Agar wo chupke se dekh raha hota, ye test fail ho jaata. **Ye tumhara proof hai ki forecast asli hai.**

Step F — Dashboard

```
.venv\Scripts\python.exe -m uvicorn server.app:app --port 8000
```

Browser: `http://127.0.0.1:8000`

Pehli baar scenario load hone mein ~50 second (saare hosts rank karta hai), uske baad har click **1 second**. Demo se pehle ek baar load kar lena.

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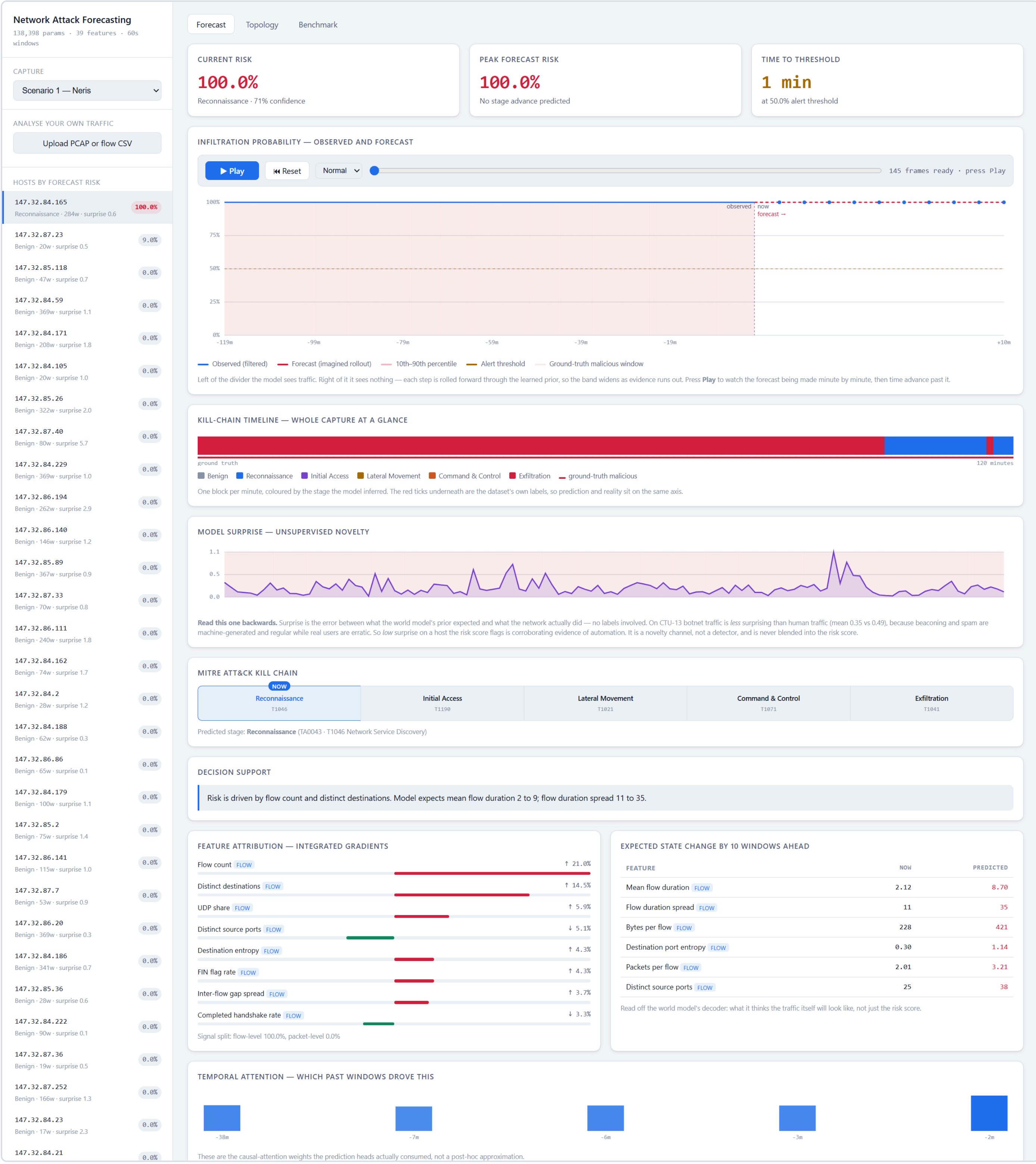
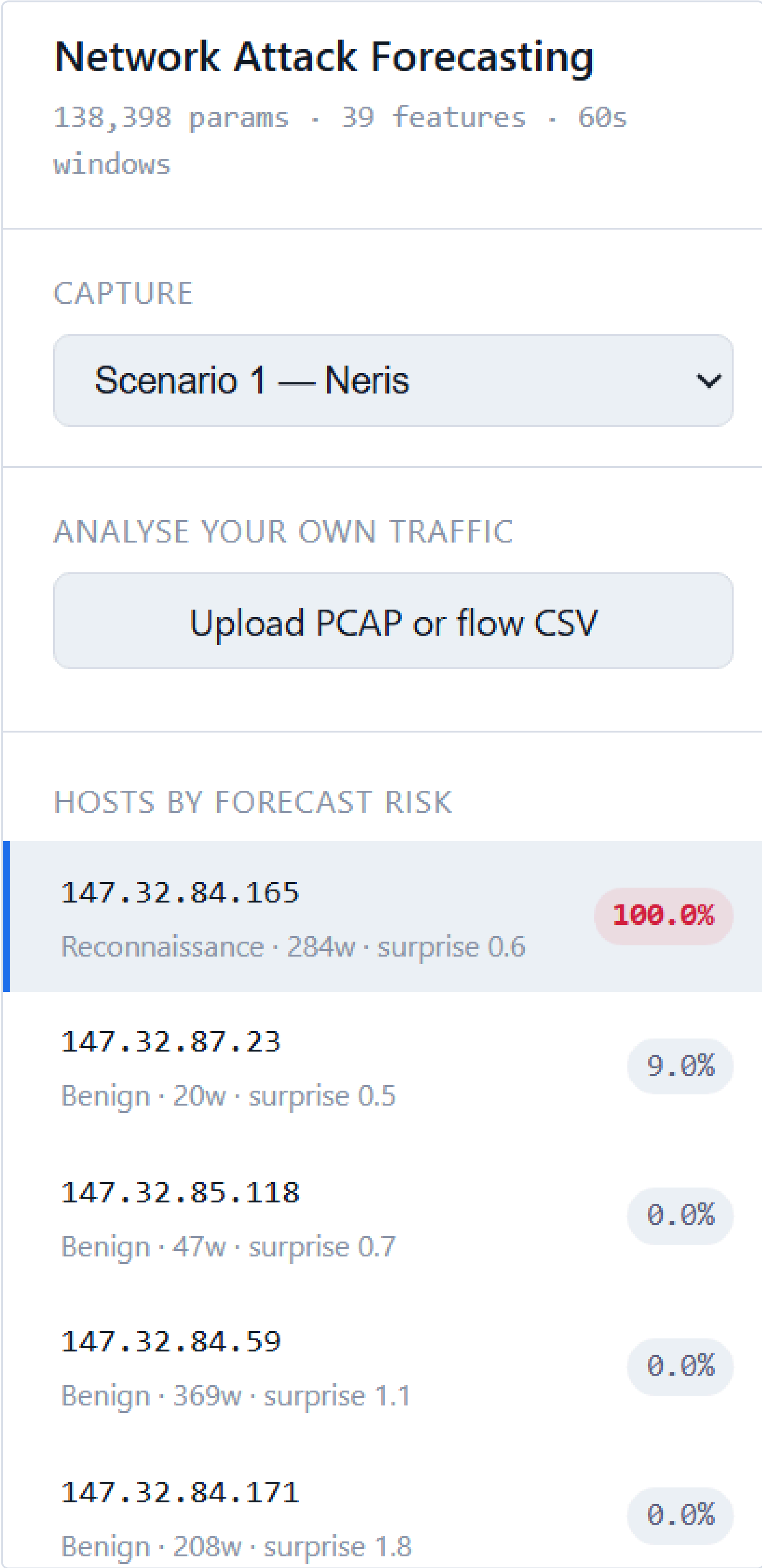
11. Dashboard mein kya-kya hai



Left side — hosts ki list



Saare computers **risk ke hisaab se sorted**. Sabse upar `147.32.84.165` — **100.0%**. Neeche waale sab **0.0%**.

Yehi wo computer hai jisme is capture mein asli malware tha. Model ne **sahi pakda**.

Beech mein — forecast chart + REPLAY (**ye sabse important hai**)

- **Baayein** divider ke — model asli traffic dekh raha hai (neeli line)
- **Daayein** divider ke — model **kuch nahi dekh raha**. Har point wo khud se bana raha hai (laal dotted line)
- **Shaded band** — 32 alag-alag rollouts ka spread. Chauda = model kam sure
- **Gulabi background** — jahan sach mein attack tha (ground truth)

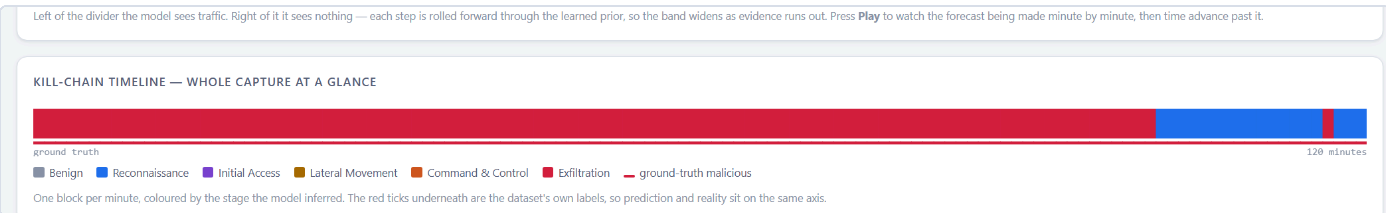
▶ **Play dabao.** Capture minute-by-minute chalega — har minute pe model ek forecast banata hai, phir waqt aage badhta hai aur pata chalta hai ki wo sahi tha ya nahi.

> Demo mein **yahi tumhara sabse strong moment hai**. Play dabao, phir kaho: "*is line ke daayein model ko traffic dikh hi nahi raha — wo imagine kar raha hai. Ab dekho waqt aage badhta hai aur pata chalta hai ki wo sahi tha.*"
>
> Static chart mein tum ye **batate** ho. Replay mein wo **dikh jaata** hai.

Slider se kisi bhi minute pe ja sakte ho, aur speed Slow/Normal/Fast chun sakte ho.

Technically: saare frames server pe **ek batched pass** mein bante hain (145 frames ≈ 2 second), har frame pe alag model call nahi hoti.

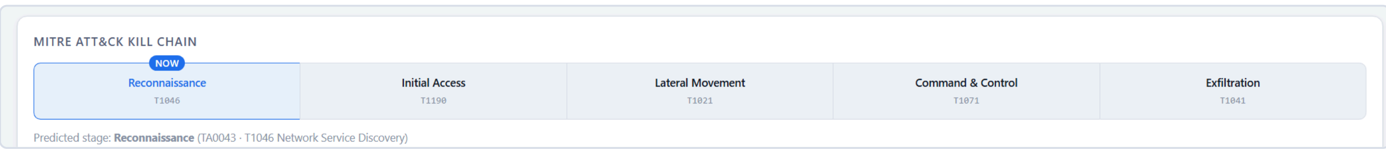
Kill-chain timeline — poora capture ek nazar mein



Har minute ka ek block, uske stage ke rang mein. **Neela = Reconnaissance, laal = Exfiltration.** Neeche laal ticks dataset ke apne labels hain.

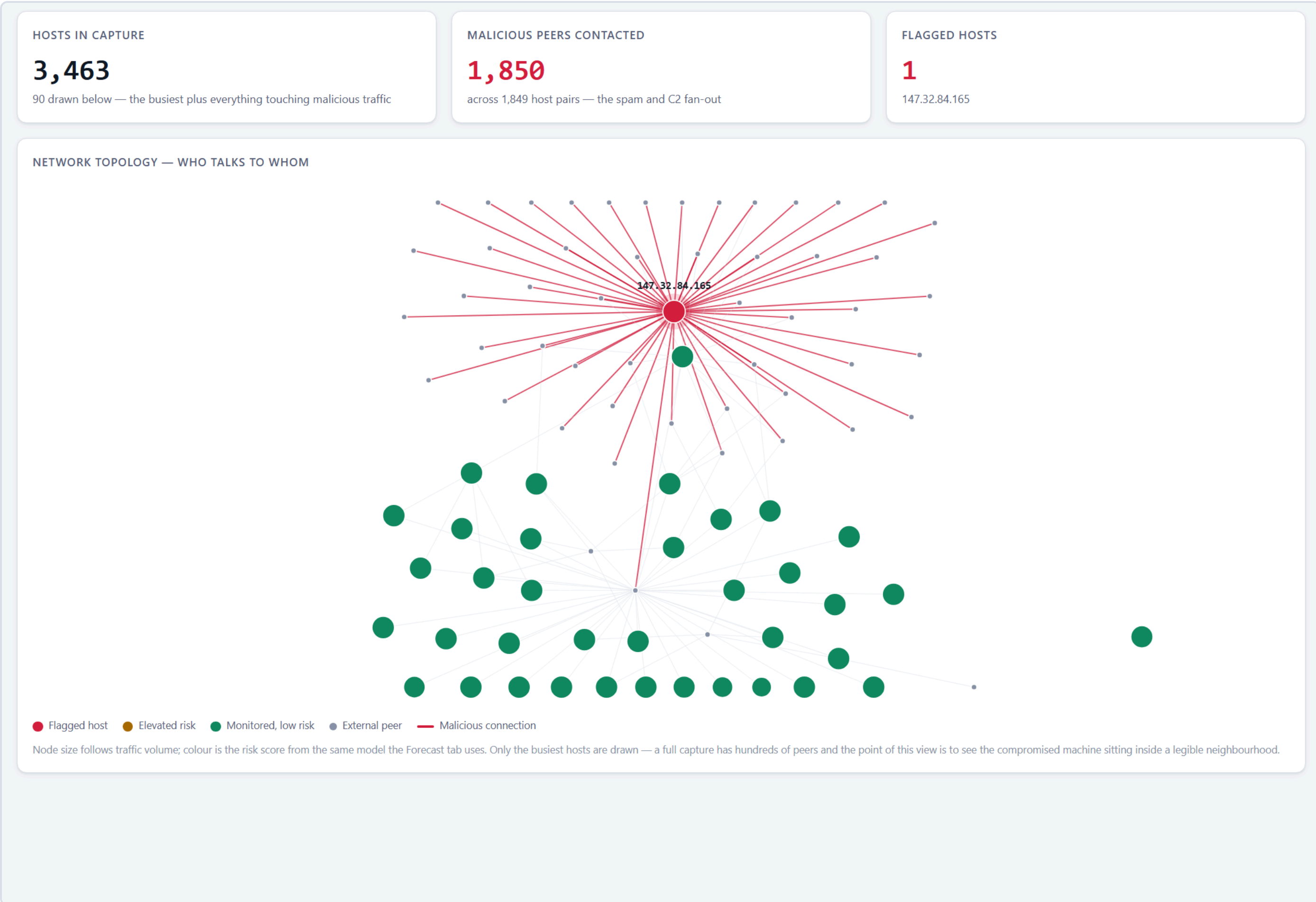
Yahi wo `111111111555555...` sequence hai jo section 6 mein thi — ab aankhon se dikhti hai. Replay chalte waqt jo hissa "dekha ja chuka" hai wo solid rehta hai, baaki halka pad jaata hai, aur ek cursor line position dikhati hai.

MITRE kill chain



5 stages, aur jahan host abhi hai wahan **NOW** ka badge. Agar model aage badhne ka predict kare to wo stage bhi highlight hoti hai.

Topology tab — network ka naksha



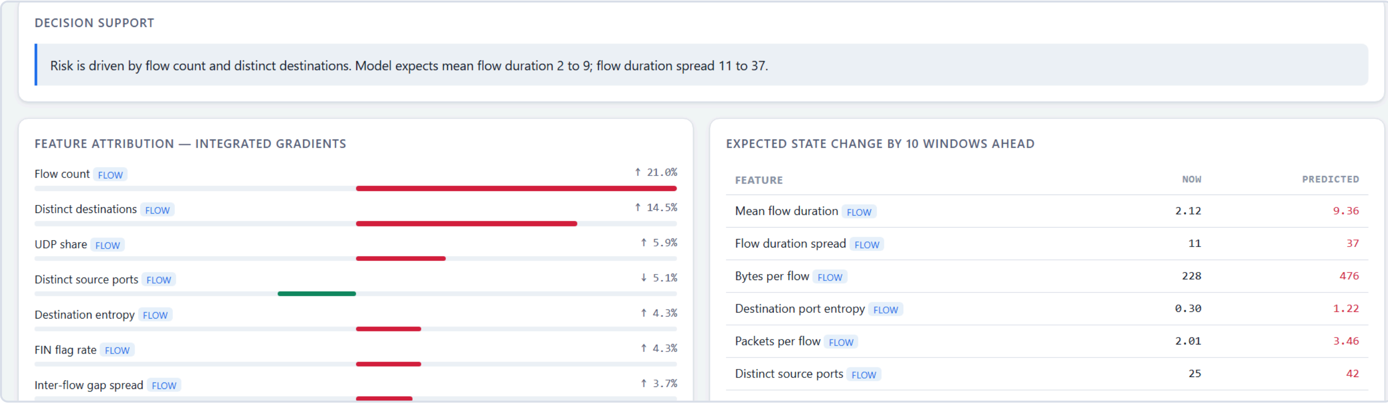
Upar **Topology** tab pe click karo.

- **Laal node** — flagged host (`147.32.84.165`)
- **Laal lines** — malicious connections
- **Hare nodes** — monitored hosts, low risk
- **Chhote grey** — external peers
- Node ka size traffic volume se

Jo laal **starburst** dikh raha hai — wo bot ka spam fan-out hai. Stats batate hain: capture mein **3,463 hosts**, aur bot ne **1,850 malicious peers** se baat ki.

> Ye slide-worthy image hai. Judge ko ek nazar mein samajh aa jaata hai ki problem kya hai — bina kisi explanation ke.

Explainability panels



- **Decision support** — ek line mein plain English wajah
- **Feature attribution** — kis number ne kitna contribute kiya (`Flow count 21.0%`)
- **Expected state change** — traffic mein kya badlega (`2.12 → 9.36`)

Upload PCAP button

Apna koi bhi capture daal sakte ho. Wo raw packets se flows reconstruct karega aur wahi forecast chalayega. Test kiya hua hai: 4 lakh packets → 18,271 flows → **12 second**.

Benchmark tab

Upar `Benchmark` pe click karo — comparison table wahan hai.

Replay andar se kaise kaam karta hai

Ye samajhna zaroori hai kyunki judge poochh sakta hai "*har frame pe model chalate ho? itna fast kaise?*"

Naive tareeka: 145 minutes = 145 API calls = 145 model runs. Bahut slow.

Humara tareeka: `observe()` ek baar chalao — wo **saare timesteps ke latent states** ek saath deta hai. Phir har timestep se rollout ko **ek batch** mein chala do:

```
observe(poora sequence)  ──▶  h[0..T], z[0..T]      ek call
                                     │
imagine(h[15..T], z[15..T])  ──▶  saare forecasts   ek batched call
```

Isliye 145 frames **2 second** mein ban jaate hain — 145 alag calls ke bajaye do calls.

Har frame ka attention history uske pehle ke **16 windows** hai (fixed length), taaki batch rectangular rahe — aur yahi length model ne training mein dekhi thi.

📁 Code: `ForecastEngine.replay()` — `src/inference.py`

Topology andar se kaise banta hai

State cells jaan-boojh kar har host ka traffic **aggregate** kar dete hain — model ko behaviour seekhna chahiye, IP addresses yaad nahi karne chahiye. Lekin isse ye pata nahi chalta ki kaun kisse baat kar raha tha.

Isliye topology ke liye ek **alag table** banti hai: `(src, dst)` pairs ke saath flow counts aur malicious flag. Wo `scenario_XX_edges.parquet` mein save hoti hai.

Ek design decision jo poochha ja sakta hai: **graph mein sirf 90 nodes kyun?** Kyunki bot ne 1,850 peers se baat ki — saare draw karne pe ek unreadable hairball banta hai (aur browser hang ho jaata hai, layout $O(n^2)$ hai). To hum priority se chunte hain: pehle flagged hosts, phir busiest malicious peers, phir volume se baaki. Aur **poora number stat card mein dikha dete hain** — wo waise bhi graph se zyada strong statement hai.

📁 Code: `build_edges()` — `src/features/windows.py`, `/api/graph` — `server/app.py`

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12. File map — kahan kya hai

```
src/
  config.py          saari settings ek jagah (window size, model size...)
  mitre.py           5 stages + CTU-13 labels se stage nikalne ka table
  features/
    flow.py          .binetflow file padhta hai, TCP flags nikalta hai
    packet.py        PCAP se packet-level features; PCAP se flows bhi banata hai
    windows.py       1-minute states banata hai  ← "39 numbers" yahan bante hain
                     build_edges() ← topology ka (src,dst) table
  dataset.py         states ko sequences mein todta hai, scaling
  model/
    world_model.py   THE MODEL — encoder, GRU, prior/posterior, heads
    baseline.py      comparison ke liye purana tareeka
  train.py           training loop
  evaluate.py        benchmark
  explain.py         3 explanation channels
  inference.py       forecast engine (dashboard isi ko call karta hai)
                     analyse() ← ek forecast | replay() ← saare frames
                     rank_hosts() ← saare hosts ek batch mein

server/
  app.py             API
  static/index.html  dashboard (poora ek file mein, offline chalta hai)

tests/
  smoke_model.py     model sahi hai ya nahi
  test_packet.py     PCAP parsing sahi hai ya nahi

docs/
  TUTORIAL.md        ye file
  architecture.md    2-page technical doc (deliverable)
  presentation.md    5-slide outline
  demo-script.md     2-min video ka script
  make_ppt.py        PPT generate karta hai
```

Kis concept ke liye kaunsi file:

Samajhna hai...	Kholo
"39 numbers" kaise bante hain	`src/features/windows.py` → `build_state_cells()`
Model kaise seekhta hai	`src/model/world_model.py` → `observe()`
Forecast kaise banta hai	`src/model/world_model.py` → `imagine()`
Stages kaise nikle	`src/mitre.py` → `_LABEL_STAGE_RULES`
Replay ke frames kaise bante hain	`src/inference.py` → `replay()`
Topology ka data kahan se aata hai	`src/features/windows.py` → `build_edges()`
Numbers kahan se aaye	`artifacts/reports/benchmark.md`

API endpoints (dashboard inhi ko call karta hai):

Endpoint	Kya deta hai
`/api/meta`	model info, available captures, MITRE stages
`/api/hosts?scenario=N`	saare hosts risk ke hisaab se ranked
`/api/analyse?scenario=N&host=X`	ek host ka forecast + explanations
`/api/replay?scenario=N&host=X`	har minute ke frames (replay ke liye)
`/api/graph?scenario=N`	network topology (nodes + edges)
`/api/upload`	PCAP/CSV upload
`/api/benchmark`	benchmark results

---

13. Cheat sheet

Agar sab bhool jao, ye 4 line yaad rakhna:

1. "*Purane tools ek connection dekh ke label dete hain. Humara model seekhta hai ki network kaise badalta hai — phir aage chala ke dekhta hai.*"
2. "*Traffic dekhna band karke bhi model aage chal sakta hai. Wahi forecast hai.*"
3. "*Purane tareeke ko wahi data diya. F1 0.984 vs 0.744.*"
4. "*Humne apne pipeline mein ek leak pakda aur hata diya.*"

Jab atak jao:

> "Wo implementation detail hai, code mein hai, main dikha sakta hu."

Phir file khol dena. Ye kamzori nahi lagti — jhoot bolne se hazaar guna behtar hai.

Jo mat bolna:

- ❌ "Ye attack hone se pehle predict karta hai" — CTU-13 pe ye defend nahi kar paoge (Finding 2 dekho). Bolo: **kill-chain progression forecast karta hai**.
- ❌ Binary F1 = 1.000 at +10 windows ko forecasting skill mat batana — target horizon pe barely change hota hai. Benchmark report mein bhi yahi likha hai.